Structural Equation Modeling (SEM) Framework for Health Workforce Investment Analysis

JKUAT - WHO SRHR Proposed Methodology version 1

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1 Introduction

This document presents an enhanced Structural Equation Modeling (SEM) framework to analyze the impact of health workforce investments, defined as both availability and productivity, on service coverage, health outcomes (DALYs), and economic productivity (GDP per capita growth). The SEM framework integrates latent constructs and observed indicators to quantify direct and indirect effects across the health system.

2 Theoretical Framework

2.1 Conceptual Model

The SEM framework includes five latent constructs:

- **Investment**: Health workforce availability and productivity
- **Efficiency**: Workforce productivity and skill-mix optimization
- **Output**: Service coverage and quality
- **Outcome**: Health outcomes measured in DALYs
- **Impact**: Economic productivity and social welfare

2.2 Hypotheses

1. H1: Health workforce investment positively affects workforce efficiency
2. H2: Workforce efficiency mediates the relationship between investment and service output
3. H3: Service output directly impacts health outcomes
4. H4: Improved health outcomes contribute to economic impact
5. H5: Direct effects exist between investment and economic impact

3 Mathematical Formulation

3.1 Structural Equation Model Specification

Let:

- η_1 = Investment (latent)
- η_2 = Efficiency (latent)
- η_3 = Output (latent)
- η_4 = Outcome (latent)
- η_5 = Impact (latent)

3.2 Structural Model Equations

$$\eta_2 = \gamma_{21}\eta_1 + \zeta_2 \tag{1}$$
$$\eta_3 = \gamma_{31}\eta_1 + \beta_{32}\eta_2 + \zeta_3 \tag{2}$$
$$\eta_4 = \beta_{43}\eta_3 + \zeta_4 \tag{3}$$
$$\eta_5 = \beta_{54}\eta_4 + \gamma_{51}\eta_1 + \zeta_5 \tag{4}$$

Where:

- γ_{ij} = direct effect from η_i to η_j
- β_{ij} = mediated effect from η_i to η_j
- ζ_i = error terms (disturbances)

3.3 Measurement Model Equations

Investment (η_1):

$$x_{11} = \lambda_{11}\eta_1 + \delta_{11} \quad \text{(Medical Doctors)}$$
$$x_{12} = \lambda_{12}\eta_1 + \delta_{12} \quad \text{(Nursing Personnel)}$$
$$x_{13} = \lambda_{13}\eta_1 + \delta_{13} \quad \text{(Midwifery Personnel)}$$
$$x_{14} = \lambda_{14}\eta_1 + \delta_{14} \quad \text{(Productivity Index)}$$

Efficiency (η_2):

$$y_{21} = \lambda_{21}\eta_2 + \epsilon_{21} \quad \text{(Skill-Mix Ratio)}$$
$$y_{22} = \lambda_{22}\eta_2 + \epsilon_{22} \quad \text{(Workload Balance)}$$
$$y_{23} = \lambda_{23}\eta_2 + \epsilon_{23} \quad \text{(Retention Rate)}$$

3.4 Total Effects Decomposition

$$\text{Total Effect}_{Investment \to Impact} = \gamma_{51} + \gamma_{21}\beta_{32}\beta_{43}\beta_{54} + \gamma_{31}\beta_{43}\beta_{54} \tag{5}$$
$$\text{Direct Effect}_{Investment \to Impact} = \gamma_{51} \tag{6}$$
$$\text{Indirect Effect}_{Investment \to Impact} = \gamma_{21}\beta_{32}\beta_{43}\beta_{54} + \gamma_{31}\beta_{43}\beta_{54} \tag{7}$$

3.5 ROI Calculation Framework

Return on Investment (ROI):

$$\text{ROI} = \frac{\text{Economic Impact (monetized)} - \text{Investment Cost}}{\text{Investment Cost}} \times 100\% \tag{8}$$

Monetized Benefits:

$$\text{Health Benefits} = \sum_{i=1}^{n} \text{DALY}_i \times VSL_i \tag{9}$$

$$\text{Economic Benefits} = \sum_{i=1}^{n} \text{Productivity Gains}_i \times W_i \tag{10}$$

$$\text{Total Benefits} = \text{Health Benefits} + \text{Economic Benefits} \tag{11}$$

Where:

- VSL = Value of Statistical Life
- W_i = Average wage rate for age group i

4 Comprehensive Observed Variables Table

Construct	**Code**	**Variable Description**
		Investment Variables
Investment	HR_1	Density per 10,000 population: Medical Doctors
	HR_2	Density per 10,000 population: Nursing Personnel
	HR_3	Density per 10,000 population: Midwifery Personnel
	HR_4	Density per 10,000 population: Dentists
	HR_5	Density per 10,000 population: Pharmacists
	HR_6	Training expenditure per health worker (USD)
	HR_7	Average salary as percentage of GDP per capita
	HR_8	Continuing education participation rate (%)
		Efficiency Variables
Efficiency	EFF_1	Skill-mix ratio (Doctors:Nurses:Midwives)
	EFF_2	Workload balance index (actual vs. optimal)
	EFF_3	Retention rate (1-year)
	EFF_4	Task-shifting effectiveness score
	EFF_5	Productivity index (patients seen per FTE)
	EFF_6	Absenteeism rate (%)
	EFF_7	Continuing medical education hours per year
		Output Variables
Output	OUT_1	Service Coverage Index: Infectious diseases
	OUT_2	Service Coverage Index: Noncommunicable diseases
	OUT_3	Service Coverage Index: Reproductive, maternal, newborn, child health
	OUT_4	Antenatal care coverage (at least 4 visits) (%)

Construct	Code	Variable Description
	OUT_5	Skilled birth attendance (%)
	OUT_6	Contraceptive prevalence rate (%)
	OUT_7	ART coverage among people living with HIV (%)
		Outcome Variables
Outcome	DALY_1	DALYs per 100,000: Communicable, maternal, perinatal, nutritional conditions
	DALY_2	DALYs per 100,000: Noncommunicable diseases
	DALY_3	DALYs per 100,000: Injuries
	DALY_4	Maternal mortality ratio (per 100,000 live births)
	DALY_5	Under-5 mortality rate (per 1,000 live births)
	DALY_6	Neonatal mortality rate (per 1,000 live births)
	DALY_7	Life expectancy at birth (years)
		Impact Variables
Impact	IMP_1	GDP per capita growth (annual %)
	IMP_2	Labor force participation rate (%)
	IMP_3	Productivity growth (annual %)
	IMP_4	Out-of-pocket health expenditure as % of total health expenditure
	IMP_5	Poverty headcount ratio at national poverty lines (%)
	IMP_6	Unemployment rate (%)
	IMP_7	Human Development Index (HDI)
		Moderator Variables
Moderators	MOD_1	Health system governance index
	MOD_2	Health financing as % of GDP
	MOD_3	Infrastructure index (facilities per capita)
	MOD_4	Supply chain management score
	MOD_5	Digital health readiness index

5 Extended SEM Diagram

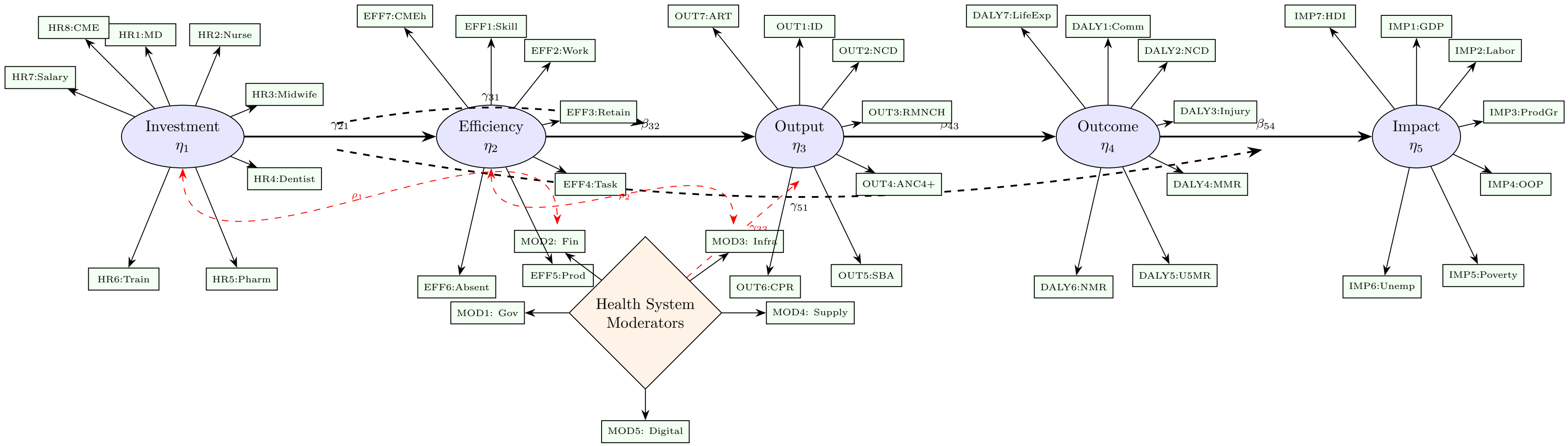

Figure 1: Complete SEM Framework with All Observed Variables (Abbreviations: MD=Medical Doctors, Nurse=Nursing Personnel, Midwife=Midwifery Personnel, Pharm=Pharmacists, Train=Training Expenditure, CME=Continuing Medical Education, Skill=Skill-Mix Ratio, Work=Workload Balance, Retain=Retention Rate, Task=Task-Shifting, Prod=Productivity, Absent=Absenteeism, CMEh=CME Hours, ID=Infectious Diseases, NCD=Non-Communicable Diseases, RMNCH=Reproductive/Maternal/Newborn/Child Health, ANC4+=Antenatal Care 4+ visits, SBA=Skilled Birth Attendance, CPR=Contraceptive Prevalence Rate, ART=Antiretroviral Therapy, Comm=Communicable Diseases, MMR=Maternal Mortality Ratio, U5MR=Under-5 Mortality, NMR=Neonatal Mortality, LifeExp=Life Expectancy, GDP=GDP Growth, Labor=Labor Participation, ProdGr=Productivity Growth, OOP=Out-of-Pocket Expenditure, Unemp=Unemployment, Gov=Governance, Fin=Financing, Infra=Infrastructure, Supply=Supply Chain, Digital=Digital Health)

6 Advanced SEM Models

6.1 Multi-Group Analysis

By Country Income Group:

$$\eta_2^{High} = \gamma_{21}^{High}\eta_1 + \zeta_2^{High} \tag{12}$$

$$\eta_2^{Low} = \gamma_{21}^{Low}\eta_1 + \zeta_2^{Low} \tag{13}$$

By Geographic Region:

$$\eta_3^{Urban} = \gamma_{31}^{Urban}\eta_1 + \beta_{32}^{Urban}\eta_2 + \zeta_3^{Urban} \tag{14}$$

$$\eta_3^{Rural} = \gamma_{31}^{Rural}\eta_1 + \beta_{32}^{Rural}\eta_2 + \zeta_3^{Rural} \tag{15}$$

6.2 Mediation Analysis

Mediation Effect:

$$\text{Mediation} = \gamma_{21} \times \beta_{32} \times \beta_{43} \times \beta_{54} \tag{16}$$

Bootstrap Confidence Intervals:

$$\text{CI}_{95\%} = [\theta_{lower}, \theta_{upper}] \tag{17}$$

6.3 Moderation Analysis

With Moderator M:

$$\eta_3 = \gamma_{31}\eta_1 + \gamma_{32}\eta_2 + \gamma_{33}M + \gamma_{34}(\eta_1 \times M) + \zeta_3 \tag{18}$$

7 Model Estimation and Evaluation

7.1 Estimation Methods

Maximum Likelihood (ML):

$$F_{ML} = \log|\Sigma(\theta)| + \text{tr}[S\Sigma^{-1}(\theta)] - \log|S| - p \tag{19}$$

Robust Estimation:

$$F_{WLS} = [s - \sigma(\theta)]'W^{-1}[s - \sigma(\theta)] \tag{20}$$

Where:

- $\Sigma(\theta)$ = model-implied covariance matrix
- S = sample covariance matrix
- p = number of observed variables

7.2 Fit Indices

Fit Index	Threshold	Interpretation
Chi-square (χ^2)	p > 0.05	Non-significant indicates good fit
CFI (Comparative Fit Index)	> 0.90	Higher values indicate better fit
TLI (Tucker-Lewis Index)	> 0.90	Incremental fit index
RMSEA (Root Mean Square Error of Approximation)	< 0.08	Lower values indicate better fit
SRMR (Standardized Root Mean Square Residual)	< 0.08	Lower values indicate better fit
GFI (Goodness of Fit Index)	> 0.90	Absolute fit index
AGFI (Adjusted GFI)	> 0.90	Adjusted for model complexity

8 ROI Calculation Examples

8.1 Scenario Analysis

Scenario 1: 10% Increase in Health Workforce

$$\begin{aligned}
\Delta\text{Investment} &= \text{Current Investment} \times 0.10 \\
\Delta\text{Output} &= \gamma_{31} \times \Delta\text{Investment} \\
\Delta\text{Outcome} &= \beta_{43} \times \Delta\text{Output} \\
\Delta\text{Impact} &= \beta_{54} \times \Delta\text{Outcome} \\
\text{ROI} &= \left(\frac{\Delta\text{Impact} - \Delta\text{Investment}}{\Delta\text{Investment}}\right) \times 100\%
\end{aligned}$$

8.2 Sensitivity Analysis

Varying Parameters:

$$\text{ROI}_{sensitivity} = f(\gamma_{21}, \beta_{32}, \beta_{43}, \beta_{54}, \gamma_{51}) \quad (21)$$

9 Policy Implications and Recommendations

9.1 Investment Prioritization

- Identify highest ROI interventions through SEM path coefficients
- Allocate resources based on marginal productivity analysis
- Consider equity implications in workforce distribution

9.2 Monitoring and Evaluation

- Establish SEM-based dashboard for workforce performance
- Implement regular fit assessment for model updating
- Use multi-group analysis for equity monitoring

10 Conclusion

The enhanced SEM framework provides a comprehensive approach to analyzing health workforce investments. By incorporating efficiency as a mediating factor and including moderating variables, the model offers nuanced insights into the pathways through which investments translate into health and economic outcomes. The ROI calculations derived from SEM coefficients enable evidence-based decision making for health workforce planning.

10.1 Further Modelling Directions

- Longitudinal SEM models for temporal analysis
- Bayesian SEM for incorporating prior knowledge
- Machine learning integration for predictive analytics
- Cross-country comparative SEM analysis